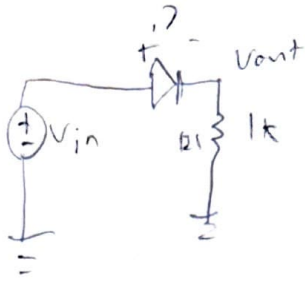
Prelab 5

Dylan Hawthorne

1.

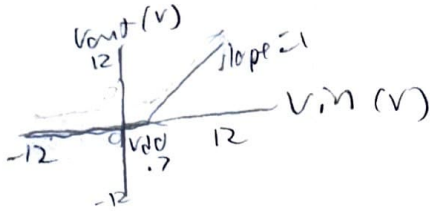


will only activate when vin > .7

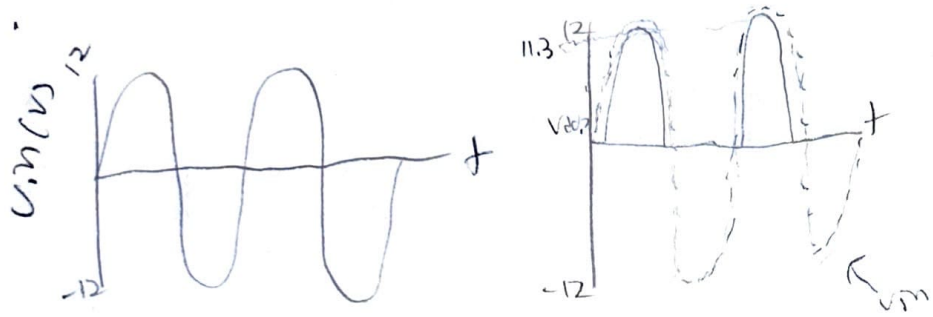
a. when vin > .7v diode is on or vin = vout

else diode is off vout = 0

b.



c.

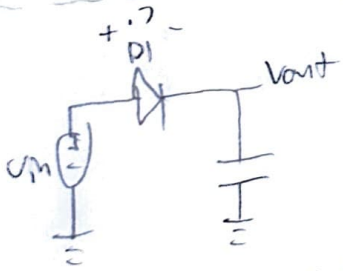


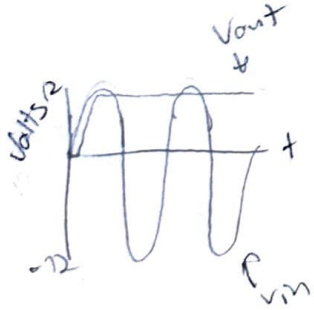
d.

$\frac{12-.7}{1000} = \boxed{11.3mA}$

e. 12 when off so $\boxed{12V}$

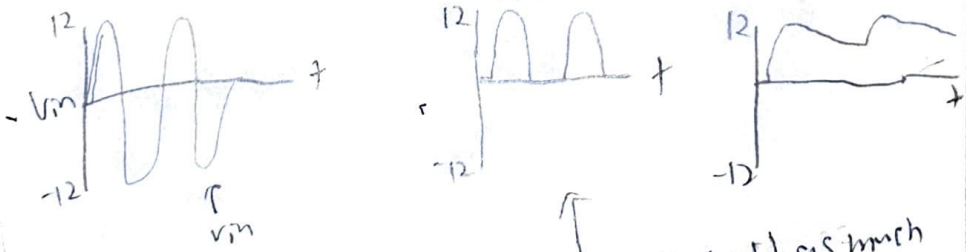
2.





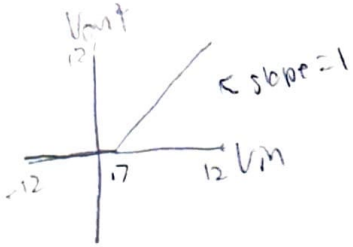
3.

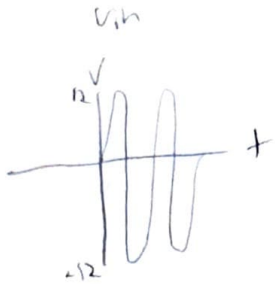
It depends on the capacitor value

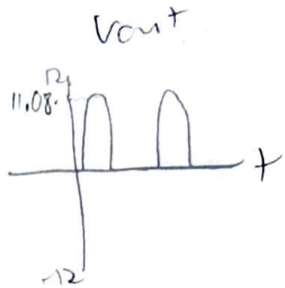


doesn't hold as much charge so it ends up behaving like a half wave rectifier.

4.







$12 - .7 = 11.3$

$11.3\left(\frac{1k}{1020}\right) = 11.08V$